

A Goldbach theorem for the Liouville function

Two commuting symmetries and a descent

1 The problem and its background

The Goldbach conjecture asks whether every even integer greater than two is a sum of two primes. One can weaken the requirement on the summands while retaining a basic feature of primality: a prime has an odd number of prime factors.

For a positive integer n , let $\Omega(n)$ be the number of its prime factors, counted with multiplicity, and put

$$\lambda(n) = (-1)^{\Omega(n)}.$$

This is the *Liouville function*. Thus $\lambda(1) = 1$, every prime has value -1 , and

$$\lambda(ab) = \lambda(a)\lambda(b), \quad \lambda(2n) = \lambda(3n) = -\lambda(n). \quad (1.1)$$

The condition $\lambda(n) = -1$ also admits many composite integers: for example, $\lambda(8) = \lambda(12) = -1$, so $20 = 8 + 12$ is an admissible sum.

In August 2018, the MathOverflow user Pablo [1] asked whether every even $N > 2$ has a representation

$$N = a + b, \quad a, b \geq 1, \quad \lambda(a) = \lambda(b) = -1. \quad (1.2)$$

Mangerel attributes the question to Shusterman [4, Problem 1.1]. Goldbach's conjecture implies this assertion. The Liouville condition retains the parity of prime factorization while allowing composite summands.

A closely related question concerns the reflected correlation

$$C(N) = \sum_{1 \leq n < N} \lambda(n)\lambda(N - n).$$

The older convolution problem of Corrádi and Kátai asks for the stronger cancellation statement $C(N) = o(N)$ as $N \rightarrow \infty$ [2]; see also the formulation in [5, Conjecture 1]. Here the issue is whether a specified sign pair occurs at every prescribed even total.

Mangerel proved, by an elementary argument, that

$$|C(N)| < N - 1 \quad \text{for every } N \geq 11, \quad (1.3)$$

and determined that equality in absolute value occurs, for $N \geq 2$, exactly at $N \in \{2, 3, 5, 10\}$ [3, Theorem 1.2]. This ensures the occurrence of both equal and unequal signs, but does not specify which equal-sign pattern occurs. In particular, it does not by itself give the pair $(-1, -1)$ in (1.2).

Mangerel subsequently proved (1.2) for every sufficiently large even integer under the Generalised Riemann Hypothesis for Dirichlet L -functions [4, Theorem 1.2]. His argument also gives, for sufficiently large prime totals, a quantitative lower bound for each of the four reflected sign patterns. A preliminary manuscript treated positive multiples of four unconditionally, using (1.3) together with elementary sign propagation [6].

There are also results averaged over the total. For example, Krishnamoorthy [5, Theorem 2] gives, for each fixed $\delta > 0$, a density-zero exceptional set outside which $|C(N)| \leq \delta N$, by classical exponential-sum and mean-square methods. Such averaged control allows exceptional totals; the theorem below concerns every even total. It does not establish the stronger asymptotic formula $C(N) = o(N)$.

The result proved here has no size threshold or analytic hypothesis.

Theorem 1.1. *For every even integer $N > 2$, there are positive integers a, b such that*

$$N = a + b \quad \text{and} \quad \lambda(a) = \lambda(b) = -1.$$

The main work goes into an assertion with the opposite sign.

Theorem 1.2. *For every prime $p > 3$, there are positive integers u, v such that*

$$2p = u + v \quad \text{and} \quad \lambda(u) = \lambda(v) = 1.$$

The usefulness of positive-positive pairs at twice a prime was already pointed out by Thomas Bloom in the original discussion [1]. Multiplication by an integer of negative Liouville sign turns such a pair into a negative-negative pair. Section 2 gives the short reduction, including the small prime factors.

The idea of the proof. Suppose a positive-positive pair is missing at $2p$. This forces one-sided reflection inequalities for λ . Extend its values below $p/2$ oddly to $\mathbb{F}_p$. The inequalities almost make multiplication by -2 and -3 preserve this extension: the failures can occur only in two short intervals. Since the two multiplications commute, these failures must satisfy a compatibility identity. The intervals leave no room for a nonzero failure.

The completed function now has exact multiplication laws for $-1, 2, 3$, and it is already multiplicative whenever a positive integer product is less than $p/2$. A descent shows that these facts force full multiplicativity. Finally, a small prime which is a quadratic residue contradicts agreement with λ .

The strategy of deriving character-like rigidity from missing sign patterns is already present in Mangerel's work [3, 4]. Commuting dilation defects also appear explicitly in [4, Section 2.2.2 and Lemma 4.3]. Here the one-sided missing-pair condition yields exact symmetries through their interval supports, and a half-interval descent completes the argument. The proof below needs neither correlation estimates nor classification of multiplicative characters. Its only classical number-theoretic input beyond elementary arithmetic is quadratic reciprocity and its supplementary laws.

2 Reduction to twice a prime

We first deduce Theorem 1.1 from Theorem 1.2. This explains the change of sign in the auxiliary problem. Only two prime factors need to be extracted.

Proof of Theorem 1.1, assuming Theorem 1.2. Write $N = 2m > 2$. If $\lambda(m) = -1$, use $N = m + m$. Otherwise $\lambda(m) = 1$. Since $m > 1$, it has a positive even number of prime factors, counted with multiplicity. Extract two of them:

$$m = rsc, \quad r, s \text{ prime}, \quad c \geq 1, \quad \lambda(c) = 1.$$

The primes r, s need not be distinct.

If one of them, say r , exceeds 3, Theorem 1.2 gives $2r = u + v$ with $\lambda(u) = \lambda(v) = 1$. Multiply by sc . Since $\lambda(sc) = -1$,

$$N = scu + scv, \quad \lambda(scu) = \lambda(scv) = -1.$$

Otherwise $r, s \in \{2, 3\}$, and $2rs \in \{8, 12, 18\}$. Use the corresponding identity

$$8 = 3 + 5, \quad 12 = 5 + 7, \quad 18 = 7 + 11,$$

and multiply both summands by c . Its positive Liouville sign preserves their negative signs. $\square$

3 A missing sum and two commuting symmetries

Fix a prime $p > 3$. Throughout this section, suppose that $2p$ has no positive-positive Liouville representation, and abbreviate $f(n) = \lambda(n)$. Only the values with $0 < n < 2p$ will be needed.

3.1 Reflection becomes an equality on the upper third

There can be no negative-negative pair summing to p : doubling both summands would give a positive-positive pair summing to $2p$. This simple observation gives all the inequalities needed below.

Lemma 3.1. *For $0 < n < p$,*

$$f(p - n) \geq -f(n), \quad f(p + n) \leq f(n). \quad (3.1)$$

For $0 < n < p/2$,

$$f(p - 2n) \geq f(n). \quad (3.2)$$

Proof. The first inequality in (3.1) says precisely that n and $p - n$ cannot both have negative sign. For the second, the only nontrivial case is $f(n) = -1$. The first inequality then gives $f(p - n) = 1$; since $(p - n) + (p + n) = 2p$, the missing positive-positive pair forces $f(p + n) = -1$. Finally, apply the first inequality at $2n$ and use $f(2n) = -f(n)$. $\square$

The inequalities are ordered as ordinary integers, with $-1 < 1$. Their useful feature is that a closed chain forces equality.

Lemma 3.2. *If $p/3 < n < p/2$, then*

$$f(n) = f(p - 2n) = -f(3n - p). \quad (3.3)$$

Proof. Put $u = p - 2n$ and $v = 3n - p$; both are positive. By (3.2), $f(n) \leq f(u)$. Since $3n = p + v$, translation in (3.1) gives $-f(n) \leq f(v)$. Finally, $p = 3u + 2v$. The two terms $3u, 2v$ cannot both have negative sign, so $f(u), f(v)$ cannot both have positive sign. Equivalently, $f(v) \leq -f(u)$. Thus

$$-f(n) \leq f(v) \leq -f(u) \leq -f(n),$$

and every inequality is an equality. $\square$

3.2 Where can the symmetries fail?

Every nonzero residue $x \in \mathbb{F}_p$ has a unique integer representative t with $-p/2 < t < p/2$. Define

$$G(0) = 0, \quad G(x) = \operatorname{sgn}(t)f(|t|) \quad (x \neq 0). \quad (3.4)$$

Thus G is odd, $G(1) = 1$, and $G(n) = f(n)$ for $0 < n < p/2$. Integer arguments of G always mean their residues modulo p . We call t the *central representative* of x .

If $0 < n < p/4$, local doubling already gives $G(-2n) = -f(2n) = f(n) = G(n)$. Lemma 3.2 gives the same identity for $p/3 < n < p/2$. Only the interval between $p/4$ and $p/3$ remains. To treat doubling and tripling together, define their *defects*

$$A(x) = G(-2x) - G(x), \quad B(x) = G(-3x) - G(x). \quad (3.5)$$

Both functions are odd. They record exactly what still needs to be proved.

Lemma 3.3. *For positive central representatives, the defects satisfy the following table. An interval in the table means its integer points.*

n	$(0, p/6)$	$(p/6, p/4)$	$(p/4, p/3)$	$(p/3, p/2)$
$A(n)$	0	0	≥ 0	0
$B(n)$	0	≥ 0	$A(n)$	0

Proof. Local doubling and Lemma 3.2 give the zeros in the first row. On the remaining interval,

$$A(n) = f(p - 2n) - f(n) \geq 0$$

by (3.2). Local tripling and Lemma 3.2 give the two outer zeros in the second row. For $p/6 < n < p/3$, the central representative of $-3n$ is $p - 3n$, and reflection at $3n$ gives

$$B(n) = f(p - 3n) - f(n) \geq 0.$$

To compare the two rows when $p/4 < n < p/3$, apply Lemma 3.2 to $u = p - 2n \in (p/3, p/2)$:

$$f(p - 2n) = -f(2(p - 3n)) = f(p - 3n).$$

Subtracting $f(n)$ gives $A(n) = B(n)$. There are no missing boundary cases, since none of $p/6, p/4, p/3, p/2$ is an integer. $\square$

3.3 The commuting square

The table leaves one extra interval on which B may be nonzero. Commutativity removes it.

Proposition 3.4. *The completion G satisfies, for every $x \in \mathbb{F}_p$,*

$$G(2x) = -G(x), \quad G(3x) = -G(x). \quad (3.6)$$

Proof. Multiplication by -2 commutes with multiplication by -3 . Taking the difference of G along the two routes from x to $6x$ gives

$$A(-3x) + B(x) = B(-2x) + A(x). \quad (3.7)$$

Indeed, both sides telescope to $G(6x) - G(x)$.

Take $p/6 < n < p/4$. The right side of (3.7) is zero: $A(n) = 0$, while $-2n$ has central magnitude $2n \in (p/3, p/2)$, where B vanishes. On the left, $-3n$ has positive central representative $p - 3n \in (p/4, p/2)$. Hence $A(-3n) \geq 0$ and $B(n) \geq 0$. Their sum is zero, so $B(n) = 0$.

The two rows of the table now agree on every positive representative. Oddness gives $A = B$ on all of $\mathbb{F}_p$. By (3.5), $G(2x) = G(3x)$, or equivalently

$$G((2/3)x) = G(x). \quad (3.8)$$

Here $2/3$ denotes $2 \cdot 3^{-1}$ in $\mathbb{F}_p$. Applying (3.8) at x and at $-2x$ shows that A , too, is invariant under multiplication by $2/3$.

Suppose A is not identically zero. The table and oddness give a positive central representative $n \in (p/4, p/3)$ with $A(n) > 0$. Let m be the central representative of $(2/3)n$. Invariance gives $A(m) = A(n) > 0$. It follows from oddness and the table that m is also positive and lies in $(p/4, p/3)$. Yet

$$3m \equiv 2n \pmod{p},$$

whereas

$$2n \in (p/2, 2p/3), \quad 3m \in (3p/4, p).$$

These are disjoint intervals inside $(0, p)$, so the congruence is impossible. Therefore $A = B = 0$. The desired identities follow from (3.5) and oddness of G . $\square$

4 From short products to full multiplicativity

The completed function already knows how to multiply whenever reduction modulo p is unnecessary. In fact,

$$G(ab) = G(a)G(b) \quad (a, b \geq 1, ab < p/2), \quad (4.1)$$

because a, b, ab are all in the interval where $G = \lambda$. We now show that exact multiplication by $-1, 2, 3$ is enough to carry this local rule through every reduction modulo p .

The arithmetic step is most transparent as a statement about cosets. For a subgroup $H \subseteq \mathbb{F}_p^\times$, a coset xH is the set obtained by multiplying x by every element of H . Throughout this section, integers appearing as arguments of a function on $\mathbb{F}_p^\times$, or in a statement of subgroup membership, are reduced modulo p .

Lemma 4.1 (Short representatives). *Let $p > 3$ be prime, and let H be a proper subgroup of $\mathbb{F}_p^\times$ containing $-1, 2, 3$. Let q be the least positive integer representing a nonzero residue outside H . Every coset of H has a positive integer representative n with*

$$n < \frac{p}{2q}.$$

Proof. First, q is prime: if it factored into two smaller positive integers, both factors would belong to H . Since $1, 2, 3, 4 \in H$, we have $q \geq 5$. Since $-1 \in H$, a missing residue has, up to sign, a missing positive representative below $p/2$; thus $q < p/2$.

Choose the smallest positive integer representative n of a fixed coset. Negation gives $n < p/2$, and n is odd: an even n could be replaced by $n/2$, because $2 \in H$. If $n = 1$, the claim follows from $q < p/2$. Hence assume $1 < n < p/2$, so $n \nmid p$. Suppose for a contradiction that $2qn \geq p$. Equality is impossible because p is odd, so $2qn > p$.

If $p < (2q - 1)n$, choose an even integer a nearest to p/n . Then

$$2 \leq a \leq 2q - 2, \quad 0 < |p - an| < n.$$

The strict inequalities follow because p/n is not an integer; in particular, it cannot lie halfway between two consecutive even integers. Write $a = 2j$, where $1 \leq j < q$. Both 2 and j belong to H , so $a \in H$. Therefore $|p - an|$ is a smaller positive representative of the same coset, a contradiction.

The endpoint $p = (2q - 1)n$ is excluded by $n \nmid p$. Only the narrow interval

$$(2q - 1)n < p < 2qn \tag{4.2}$$

remains. Since the prime $q \geq 5$ is not divisible by 3, exactly one of $2q - 1$ and $2q + 1$ is divisible by 3. Choose that integer as a . It has the form $a = 3j$ with $1 \leq j < q$, so once again $a \in H$. The bounds (4.2) give

$$0 < |p - an| < 2n.$$

This remainder might be larger than n , but it is even: a, n, p are all odd. Consequently

$$r = \frac{|p - an|}{2} \tag{4.3}$$

is a positive integer smaller than n . Its residue is $\pm(a/2)n$, which lies in the same coset because $-1, a, 2 \in H$. This final contradiction proves the lemma. $\square$

The last step is the reason for using both small multipliers. Multiples of 2 handle all but the interval (4.2); a nearby multiple of 3, followed by division by 2, handles the interval that remains.

Proposition 4.2 (Half-interval extension). *Let $p > 3$ be prime, and let $F : \mathbb{F}_p^\times \rightarrow \{\pm 1\}$ satisfy $F(1) = 1$. Suppose that*

$$F(cx) = F(c)F(x) \quad \text{for } c \in \{-1, 2, 3\}, x \in \mathbb{F}_p^\times, \tag{4.4}$$

$$F(ab) = F(a)F(b) \quad \text{for } a, b \in \mathbb{Z}_{\geq 1}, ab < p/2. \tag{4.5}$$

Then $F(xy) = F(x)F(y)$ for every $x, y \in \mathbb{F}_p^\times$.

Proof. Call a residue $c \in \mathbb{F}_p^\times$ *exact* if $F(cx) = F(c)F(x)$ for every $x \in \mathbb{F}_p^\times$, and let H be the set of exact residues. It contains 1 and is closed under multiplication, by composing the two multiplication laws. It is also closed under inverses: exactness at c , applied first to c^{-1} and then to $c^{-1}x$, gives $F(c^{-1}x) = F(c^{-1})F(x)$. Thus H is a subgroup containing $-1, 2, 3$.

Suppose H is proper, and let q be its least missing positive integer. The multiplicative defect

$$D(x) = \frac{F(qx)}{F(q)F(x)}$$

is constant on each coset of H . Indeed, for $c \in H$, exactness at c cancels the same factor $F(c)$ from numerator and denominator, giving $D(cx) = D(x)$.

By Lemma 4.1, every coset contains an integer $n < p/(2q)$. Then $qn < p/2$, so (4.5) gives $D(n) = 1$. Hence $D = 1$ everywhere, which means that q is exact after all. Therefore $H = \mathbb{F}_p^\times$. $\square$

Corollary 4.3. *Under the missing-pair assumption of Section 3, the completion G is multiplicative on $\mathbb{F}_p^\times$.*

Proof. Oddness gives the exact multiplier -1 . Evaluating (3.6) at $x = 1$ gives $G(2) = G(3) = -1$, so the same identities give exact multiplication by 2 and 3. Equation (4.1) supplies short multiplicativity. Apply Proposition 4.2. $\square$

5 A small prime quadratic residue

Every multiplicative sign function takes the value 1 on nonzero squares. A single prime which is a square modulo p will therefore contradict agreement with λ . The half interval is large enough to contain such a prime whenever the sign at -1 is negative.

We use the usual quadratic-residue criteria and quadratic reciprocity; see, for example, Ireland and Rosen [7, Chapter 5]. The prime-divisor observation in the next lemma is standard; it also appears in the official HMMT 2020 solutions [8, Problem 9]. For the classical study of the least prime quadratic residue, see Chowla, Cowles and Cowles [9].

Lemma 5.1. *If $p > 3$ is prime and $p \equiv 3 \pmod{4}$, then there is a prime $\ell < p/2$ which is a quadratic residue modulo p . In fact, every prime divisor of $(p+1)/4$ has this property.*

Proof. The integer $L = (p+1)/4$ is greater than one. Choose a prime $\ell \mid L$; then $\ell \leq L < p/2$. If $\ell = 2$, then L is even and $p \equiv 7 \pmod{8}$, so 2 is a quadratic residue modulo p .

If ℓ is odd, quadratic reciprocity and $p \equiv 3 \pmod{4}$ give

$$\left(\frac{\ell}{p}\right) = (-1)^{(\ell-1)/2} \left(\frac{p}{\ell}\right) = \left(\frac{-p}{\ell}\right) = \left(\frac{1}{\ell}\right) = 1,$$

since $\ell \mid p+1$. Here the parentheses denote the Legendre symbol. Thus ℓ is a nonzero square modulo p in either case. $\square$

Proof of Theorem 1.2. Suppose the asserted pair is missing. The preceding sections construct an odd multiplicative function $G : \mathbb{F}_p^\times \rightarrow \{\pm 1\}$ agreeing with λ on $0 < n < p/2$.

For every nonzero x ,

$$G(x^2) = G(x)^2 = 1.$$

But $G(-1) = -1$, so -1 is not a square modulo p . The standard criterion gives $p \equiv 3 \pmod{4}$. Choose the prime ℓ from Lemma 5.1. Since ℓ is a nonzero square and $\ell < p/2$,

$$1 = G(\ell) = \lambda(\ell) = -1,$$

a contradiction. $\square$

A The accompanying formal proof

Theorem 1.1 is formalized in Lean 4 [10] with mathlib [11]. The final declaration is

```
LiouvilleGoldbach.liouville_goldbach.
```

It has exactly the stated hypotheses and conclusion, and uses mathlib’s Liouville function on natural numbers.

The principal steps are in `Commuting.lean`, `Descent.lean`, `Coset.lean`, and `Residue.lean`. The third module proves the short-representative lemma in a more general invariant-set form and deduces the same extension theorem. `Reduction.lean` and `Main.lean` complete the argument.

The source archive pins Lean to v4.34.0-rc2 and mathlib to commit

```
de2ef68216c6074f338c8e61890ee0a379ddfb9b.
```

The complete build passes. The final axiom audit reports only `propext`, `Classical.choice`, and `Quot.sound`, the standard logical foundations used by Lean and mathlib. No unfinished proofs or additional mathematical axioms occur. A dependency audit also checks that the final theorem uses the new commuting and descent arguments. The accompanying archive provides the source, reproduction instructions, and all verification logs.

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